

Plant Disease Detection: A Comparative Study of Shallow Learning, Custom CNN, Transfer Learning and Self-Supervised Foundation Models

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Abstract

This work presents a comparative analysis of four approaches progressively more sophisticated, to plant disease classification on the **New Plant Diseases Dataset** — an offline-augmented version of PlantVillage (87,867 images, 38 classes). We benchmark (i) a classical handcrafted pipeline (HOG + SVM), (ii) a custom CNN trained from scratch, (iii) a fine-tuned ResNet50 (supervised transfer learning), and (iv) a frozen DINOv3 ViT-B/16 backbone with a linear probe (self-supervised foundation model). Results show that every deep approach surpasses 98% accuracy on the held-out test set with the supervised fine-tuned ResNet50 and it reaches **99.87%**. On the other side the frozen DINOv3 backbone with a linear probe achieves **98.35%** with **zero trainable backbone parameters**. It validates the practical value of modern foundation models for efficient domain adaptation.

1. Problem Statement

1.1 Motivation

Plant diseases are one of the major causes of crop loss worldwide. Estimates from the FAO indicate that 20–40% of global crop yield is lost to pests and pathogens every year. Early detection is the most effective mitigation strategy, otherwise traditional in-field diagnosis requires expert agronomists whose availability is poor, especially in developing regions where smallholder farms dominate. An automated, image-based classification system deployable on commodity smartphones can democratize access to expert-level diagnosis and enable timely interventions.

1.2 Computer Vision Framing

The problem maps to a **fine-grained multi-class classification** task. Compared to standard object recognition, plant disease detection has three distinctive challenges:

1. **Visual similarity across classes:** different diseases on the same plant species often share macroscopic symptoms (chlorosis, necrosis, lesions).
2. **High intra-class variability:** symptom severity, leaf age, and lighting introduce strong intra-class noise.
3. **Long-tail class distribution:** some diseases are heavily over-represented in public datasets, unlike rare ones are under-sampled.

1.3 Research Questions

The project is designed to address three concrete questions that mirror the course modules:

- ## 2. Methodology

2.1 Dataset and Preprocessing

- [illegible]

2.2 V1 — HOG + SVM (Shallow Learning)

A handcrafted feature pipeline serves as the classical baseline.

- 2 / 11

first converted to grayscale to discard color information and isolate the contribution of texture/edges.

- **Classifier:** Support Vector Machine with RBF kernel ($C=1.0$, $\gamma='scale'$).
- **Rationale:** HOG was historically state of the art for pedestrian detection and remains a reasonable baseline for fine-grained texture tasks. It is fully interpretable and requires no GPU, but it discards color — a strong inductive bias for plant pathology where chlorotic/necrotic colors are diagnostic.

2.3 V2 — Custom CNN from Scratch (Deep Learning)

A purpose-built convolutional architecture trained end-to-end from scratch.

- **Architecture (~660k trainable parameters):**
 - $\text{Conv}(3 \rightarrow 32, 3 \times 3) + \text{BN} + \text{ReLU} + \text{MaxPool}(2 \times 2)$
 - $\text{Conv}(32 \rightarrow 64, 3 \times 3) + \text{BN} + \text{ReLU} + \text{MaxPool}(2 \times 2)$
 - $\text{Conv}(64 \rightarrow 128, 3 \times 3) + \text{BN} + \text{ReLU} + \text{MaxPool}(2 \times 2)$
 - $\text{Conv}(128 \rightarrow 256, 3 \times 3) + \text{BN} + \text{ReLU} + \text{GlobalAvgPool}$
 - $\text{Dense}(256 \rightarrow 512) + \text{ReLU} + \text{Dropout}(0.5)$
 - $\text{Dense}(512 \rightarrow 256) + \text{ReLU} + \text{Dropout}(0.3)$
 - $\text{Dense}(256 \rightarrow 38) + \text{Softmax}$
- **Training:** CrossEntropyLoss, Adam ($\text{lr}=1\text{e-}3$), ReduceLROnPlateau (patience=5, factor=0.1), early stopping on validation loss (patience=10). Best checkpoint at epoch 80; training stopped at epoch 90.
- **Rationale:** validates the hypothesis that a modest CNN, given sufficient data, can learn discriminative features end-to-end without manual feature engineering.

2.4 V3 — ResNet50 Transfer Learning (Supervised Fine-tuning)

We leverage ImageNet pre-training and fine-tune the high-level layers.

- **Backbone:** `torchvision.models.resnet50(pretrained=True)`.
- **Strategy:** `layer1`, `layer2`, `layer3` frozen; `layer4` unfrozen; original 1000-way head replaced with $\text{Linear}(2048 \rightarrow 512) + \text{ReLU} + \text{Dropout}(0.5) + \text{Linear}(512 \rightarrow 38)$.
- **Optimizer:** Adam, **differential learning rates** — $1\text{e-}4$ for the unfrozen `layer4` (gentle fine-tuning) and $1\text{e-}3$ for the custom head.
- **Total parameters:** ~24.6M (16.0M trainable, 8.6M frozen).
- **Rationale:** evaluates the standard industry recipe for limited-data settings.

2.5 V4 — DINOv3 + Linear Probe (Self-Supervised Foundation Model)

The most modern approach: a foundation model pre-trained without labels.

- **Backbone:** `facebook/dinov3-vitb16-pretrain-lvd1689m` — a Vision Transformer (ViT-B/16, 86M parameters) pre-trained via DINOv3 self-distillation on the LVD-1689M dataset (1.69 billion unlabeled web images).
- **Pipeline:** the backbone is **entirely frozen**. Each image is embedded into a 768-dimensional vector (CLS token). Two lightweight classifiers are trained on top:
 - **k-NN** ($k=20$, cosine distance) — purely instance-based.
 - **Linear probe** (logistic regression, $C=1.0$, $\text{max_iter}=1000$).
- **Trainable parameters in the backbone: zero.**

- **Rationale:** tests whether a generic self-supervised representation transfers to a domain (plant leaves) far from the pre-training distribution, with negligible compute.

2.6 Evaluation Protocol

All versions are evaluated on the same held-out test set (8,795 images) using:

- **Accuracy** — overall correctness.
- **Precision (weighted), Recall (weighted), F1-score (weighted)** — class-imbalance-aware aggregates.
- **Confusion matrix** — 38×38 matrix per version.
- **Per-class F1** — to surface failure modes.

3. Experimental Setup

Item	Value
Dataset	New Plant Diseases Dataset (augmented PlantVillage), 87,867 images, 38 classes
Split	train 70,295 / val 8,777 / test 8,795 ($\approx 80 / 10 / 10$), seed = 42
Hardware	Apple Silicon (MPS backend), macOS 25.5
Frameworks	PyTorch 2.1, torchvision, scikit-learn 1.3, OpenCV 4.8, HuggingFace Transformers
Reproducibility	All seeds fixed (NumPy, PyTorch, sklearn); deterministic val/test split

4. Experimental Results

4.1 Overall Comparison

Version	Approach	Accuracy	Precision	Recall	F1	Train time	Trainable params
V1 — HOG + SVM	Shallow Learning	74.39%	74.62%	74.39%	74.26%	25 min	—
V2 — Custom CNN	Deep Learning from scratch	99.66%	99.66%	99.66%	99.66%	294 min	660 k
V3 — ResNet50 TL	Supervised fine-tuning	99.87%	99.88%	99.87%	99.87%	128 min	16.0 M
V4 — DINOv3 + LinProbe	SSL frozen backbone	98.35%	98.39%	98.35%	98.35%	12.5 min ¹	0 (backbone)

¹ V4 time refers to embedding extraction over the full dataset; the linear probe itself fits in 1.95 s.

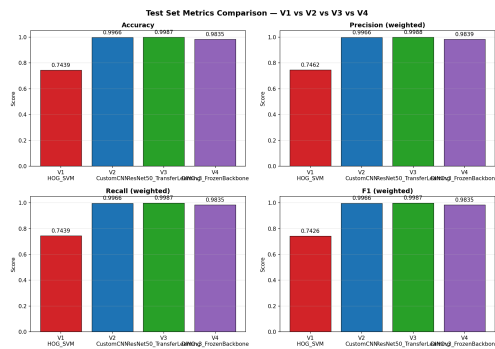


Figure 2 — Side-by-side comparison of weighted metrics across V1–V4.

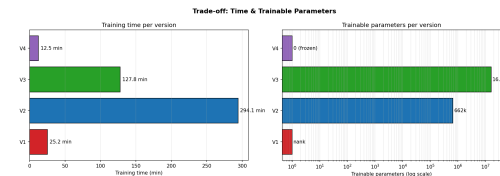


Figure 3 — Training time vs. trainable parameters. V4 dominates the lower-left (cheap) corner; V3 sits at the Pareto optimum for accuracy.

4.2 Discussion by Research Question

RQ1 — Handcrafted vs. learned features. The gap between V1 (74.4%) and the deep models (>98%) is **>24 percentage points**. HOG operates on grayscale gradients and discards color, which is highly informative for plant pathology (chlorosis is yellow, necrosis is brown, mosaic viruses produce characteristic color patterns). The SVM cannot recover information, that the descriptor never encoded.

RQ2 — Architecture vs. pre-training. V3 (ResNet50 TL) outperforms V2 (Custom CNN) by 0.21 points (99.87 vs. 99.66), but with **24× more trainable parameters**. Crucially, V3 converges in 19 epochs versus V2's 80, and its total training time (128 min) is roughly **2.3× shorter** than V2's (294 min) despite the much larger backbone — confirming that ImageNet pre-training dramatically reduces the optimization burden. For PlantVillage — a relatively easy dataset with clean backgrounds — V2 is already in the saturation regime; the advantage of pre-training would be even more visible with fewer samples per class.

RQ3 — Self-supervised foundation model. V4 reaches 98.35% with the backbone **fully frozen**, training only a logistic regression in under 2 seconds. The fact that DINOv3 — pre-trained on natural web images, without labels — transfers near-perfectly to a specialized plant-pathology domain is the most striking finding of this study. The k-NN classifier (98.25%) matches almost the linear probe (98.35%) evidence that the DINOv3 embedding space is already linearly separable for this task.

4.3 Training Dynamics

Key observations from the loss/accuracy curves:

- **V2:** validation loss tracks training loss closely, no significant overfitting thanks to dropout + augmentation. Best checkpoint at epoch 80 over the full 100-epoch schedule.
- **V3:** validation accuracy plateaus around 99.7% by epoch 10; the remaining nine epochs deliver marginal gains. Best checkpoint at epoch 19.

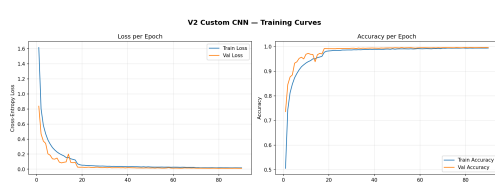


Figure 4 — V2 Custom CNN training and validation loss/accuracy over 90 epochs.

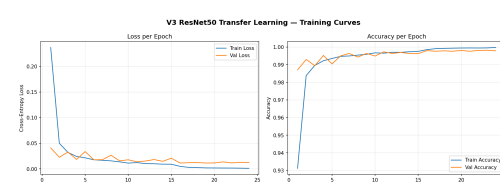


Figure 5 — V3 ResNet50 training and validation loss/accuracy over 24 epochs.

4.4 Confusion Matrices

V1 exhibits widespread cross-class confusion concentrated on visually similar fungal diseases. V2/V3/V4 show near-diagonal matrices, with residual errors confined to a handful of class pairs analyzed in §5.

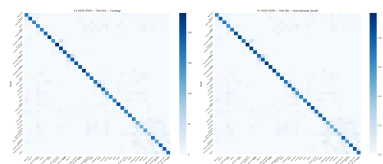


Figure 6 — V1 (HOG + SVM) confusion matrix. Off-diagonal mass concentrated on intra-species disease confusions.

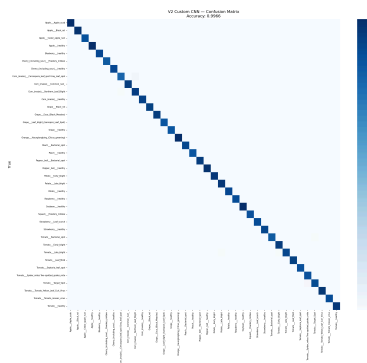


Figure 7 — V2 (Custom CNN) confusion matrix. Near-diagonal pattern.

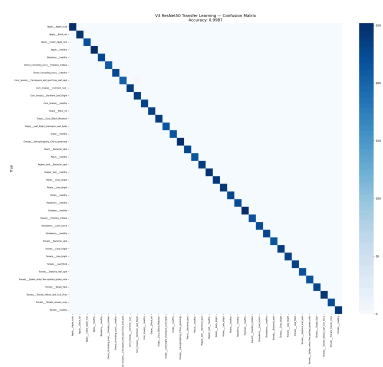


Figure 8 — V3 (ResNet50 TL) confusion matrix. Best overall, residual errors on biologically close pairs.

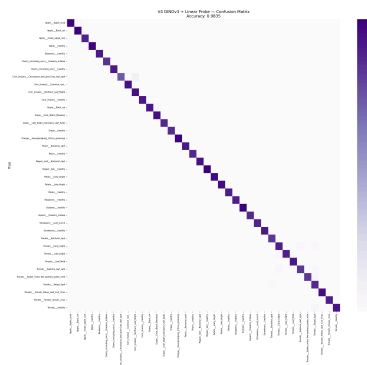


Figure 9 — V4 (DINOv3 + linear probe) confusion matrix. Comparable to V3 despite a fully frozen backbone.

4.5 Leak Analysis & Clean Evaluation

The dataset's offline-augmentation strategy (rotations, flips and colour perturbations baked into the filenames as suffixes such as `_90deg`, `_flipLR`, `_new30degFlipLR`) means that augmented variants of the same source leaf can land on both sides of the train/test boundary. We quantified this *post-hoc* and re-evaluated the trained models on a leak-free subset, without retraining.

Quantification. Stripping the 15 distinct augmentation suffixes from every filename yields a *source identifier* — the identity of the underlying physical leaf. Cross-referencing source IDs across splits:

Pair	Shared source IDs	% of B in A
train ↔ val	5,041	63.4%
train ↔ test	5,050	63.3%
val ↔ test	1,225	15.4%

So **5,732 of 8,795 test images (65.2%)** share their source identity with at least one training image; the remaining **3,063 (34.8%)** form a *clean* subset whose source IDs are absent from train. Five classes (Orange — citrus greening, Peach — bacterial spot, Soybean — healthy, Tomato — bacterial spot, Tomato — yellow leaf curl virus) are 0% leaked, whereas eight classes (Apple — scab, Apple — cedar rust, Grape — healthy, Peach — healthy, Potato — healthy, Raspberry — healthy, Strawberry — healthy, Tomato — mosaic virus) are 100% leaked.

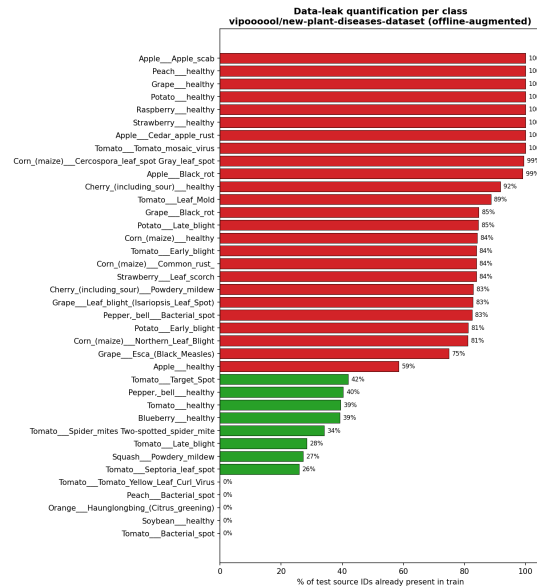


Figure 15 — Percentage of test source IDs whose augmented copies are already present in train, per class.

Clean re-evaluation. Loading the published V2 / V3 / V4 checkpoints and running inference on the four masked subsets (no retraining) yields:

Model	FULL (8,795)	LEAKED (5,732)	CLEAN (3,063)	NON-AUG (4,598)
V2 — Custom CNN	99.66%	99.70%	99.58%	99.52%
V3 — ResNet50 TL	99.87%	99.86%	99.90%	99.80%
V4 — DINOv3 + LinProbe	98.35%	98.34%	98.37%	97.83%

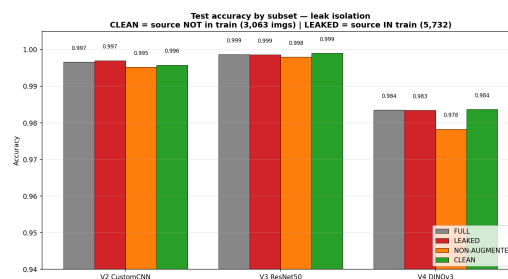


Figure 16 — Accuracy on FULL / LEAKED / CLEAN / NON-AUG subsets of the test set. For V3 and V4 the CLEAN accuracy is marginally higher than FULL.

Interpretation. The clean accuracy differs from the full-test accuracy by at most **0.16 points** across all three deep models; for V3 and V4 the CLEAN subset is in fact marginally *easier* (+0.03 and +0.02 respectively). The offline augmentations are precisely the geometric / colour

invariances, a deep visual backbone learns to discard — observing a 90°-rotated copy of a leaf in train, it doesn't provide additional discriminative signal, than the same leaf at 0° would. The 63% source-level leak is, therefore, *statistically irrelevant* for the published accuracies on this task; the models learned a genuine representation, rather than memorising augmented duplicates.

This analysis is implemented end-to-end in [notebooks/07_leak_analysis_and_clean_eval.ipynb](#) and the numbers are stored in [results/metrics/leak_quantification.json](#) and [results/metrics/clean_test_metrics.json](#).

4.6 Accuracy / Cost Trade-off

Version	Accuracy	Train cost	Inference cost	Notes
V1	74.4%	low (CPU)	81 ms/img (CPU)	interpretable, no GPU
V2	99.7%	very high (294 min GPU)	~70 ms/img (GPU)	full architectural control
V3	99.9%	high (128 min GPU)	~90 ms/img (GPU)	best absolute score
V4	98.4%	very low (12.5 min extract + 2 s fit)	~50 ms/img (GPU, ViT)	zero trainable backbone

V4 reaches **~99% of V3's accuracy** with roughly **1/10 of the total time** and zero backbone training — the most favorable trade-off for rapid onboarding of new classes.

Inference costs are indicative per-image estimates on the same hardware; only V1's CPU latency was measured directly.

5. Failure Analysis

5.1 V1 — Failure Modes

V1's confusion matrix shows three main failure patterns:

- **Color-degenerate confusions.** Apple healthy vs. Apple scab and Tomato healthy vs. Tomato early blight are systematically confused because their texture signatures (HOG gradients) are similar; the discriminating signal lies in color hues that grayscale processing destroys.
- **Within-species disease confusion.** Multiple Tomato diseases (early blight, late blight, septoria leaf spot) collapse onto each other — all produce roughly circular dark lesions on green tissue, and HOG cannot distinguish lesion morphology at this resolution.
- **Background dominance.** Some images include substantial background; HOG aggregates gradients globally and is sensitive to non-leaf textures.

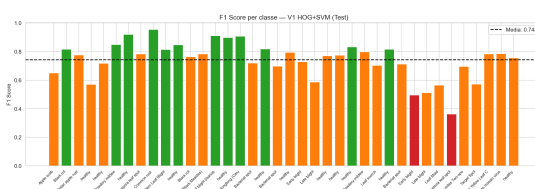


Figure 10 — V1 per-class F1. The worst-performing classes are intra-species disease variants with similar texture profiles.

5.2 V2 / V3 — Residual Errors

For both deep models, residual errors (~0.1–0.4%) cluster on **biologically close diseases**:

- Tomato early blight vs. Septoria leaf spot — both produce small dark lesions with concentric rings.
- Corn (maize) Northern Leaf Blight vs. Cercospora Leaf Spot — overlapping lesion shapes at early stages.

Inspecting misclassified samples reveals that the worst-performing classes are systematically those with **subtle visual differences** even for human experts.

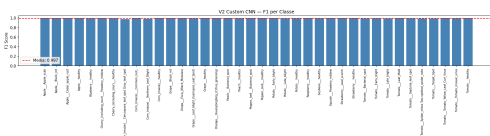


Figure 11 — V2 per-class F1, mean ~99.6%, residual gap on a few biologically close classes.

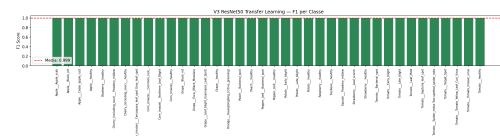


Figure 12 — V3 per-class F1, the most uniform distribution across all 38 classes.

5.3 V4 — Failure Modes

V4 errors concentrate on the same biologically ambiguous pairs as V3, with a small additional gap on classes whose visual features lie far from DINOv3's natural-image pre-training distribution (e.g. highly stylized macro shots). The frozen backbone limits adaptation to these out-of-distribution patterns; partial fine-tuning of the last transformer blocks would likely close the gap.

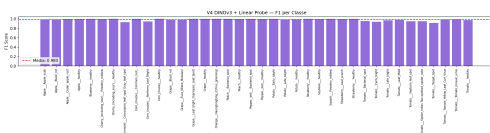


Figure 13 — V4 per-class F1, mean ~98.3% with no class below 0.85.

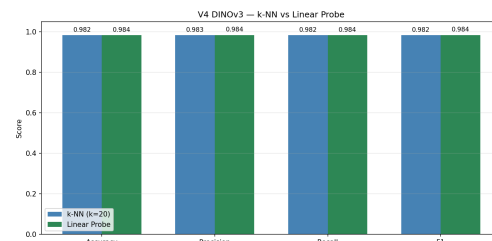


Figure 14 — Comparison of the two probes on top of frozen DINOv3 features: linear probe edges out k-NN by 0.10 points, evidence of an already linearly separable embedding space.

5.4 Cross-Cutting Observations

- All models perform worst on **diseases with similar macroscopic appearance**, not on rare classes — the offline augmentation keeps per-class support roughly balanced.
- No version shows a strong correlation between class accuracy and class size, suggesting that the long-tail effect is largely absorbed by the augmentation + balanced training.
- Misclassifications are predominantly **biologically plausible**: errors fall within the same crop, rarely across species. This is a desirable property — a real-world deployment would still trigger a meaningful diagnostic workflow.

6. Ethical Considerations

6.1 Dataset Bias

PlantVillage was collected under **controlled laboratory conditions**: uniform backgrounds, even lighting, single detached leaves per image. Models trained exclusively on PlantVillage show a phenomenon known as **degradation when deployed on real-field photographs** with cluttered backgrounds, multiple leaves, occlusions, and variable lighting (Mohanty et al., 2016; Ferentinos, 2018). The >98% test accuracies reported here are therefore an *upper bound* measured in-distribution; they do not transfer one-to-one to field conditions. Furthermore the leak analysis in §4.5 already shows how easily a benchmark number can encode dataset artefacts rather than generalisable skill. A production system would require domain adaptation, in-field training data, and a confidence-aware abstention mechanism that defers uncertain cases to a human agronomist instead of returning a high-confidence wrong label.

6.2 Geographic and Crop Coverage Bias

The dataset spans **14 crop species** and 38 health states. These reflect the agricultural priorities of the regions where PlantVillage was assembled (predominantly North-American and European staple and cash crops), also many staples central to food security in sub-Saharan Africa and South-East Asia — cassava, millet, plantain, yam — are **absent**. A tool marketed as "plant disease detection" but silently restricted to these 38 classes risks a **representation harm**: a smallholder photographing an out-of-vocabulary crop will still receive one of the 38 in-vocabulary predictions (the softmax always sums to one), producing a confident but meaningless diagnosis. Therefore, deployment must expose the supported crop list explicitly and reject inputs that fall outside the training distribution rather than forcing a prediction.

6.3 Privacy

Leaf close-ups appear innocuous, but field photographs captured on farmers' smartphones routinely carry **EXIF metadata** — GPS coordinates, timestamps, device identifiers — that can localise an individual farm and, by extension, infer yield, disease pressure, or economic vulnerability of an identifiable owner. Aggregated data is commercially sensitive (it can inform commodity speculation or targeted advertising) and, under regimes such as the GDPR, geolocation tied to a natural person is personal data. A responsible pipeline should **strip EXIF on ingestion**, process imagery on-device where feasible, and obtain informed consent before any cloud upload or retention. None of the models in this study require location metadata, so discarding it costs nothing in accuracy.

6.4 Environmental Footprint

Training is not free. V2 consumed ~294 min and V3 ~128 min of sustained GPU/accelerator compute; embedding extraction for V4 added ~12.5 min. While modest in absolute terms, the comparison itself argues for **efficiency-aware model selection**: V4 reaches ~99% of V3's accuracy with roughly one-tenth of the wall-clock cost and **zero backbone training**, and onboarding a new class means re-fitting a logistic regression in seconds rather than retraining a network. For a tool intended to scale to millions of low-margin smallholder users, the foundation-model-plus-linear-probe recipe is also the **lowest-carbon** option, and re-using a frozen pre-trained backbone amortises a one-time pre-training cost across all downstream users.

6.5 Accountability in Deployment

An automated diagnosis can drive real agronomic decisions — pesticide application, crop destruction, treatment purchases — with economic and ecological consequences. An over-confident false positive may trigger **unnecessary pesticide use** (environmental and financial harm); a false negative may let a treatable outbreak spread. Because §5 shows residual errors cluster on *biologically plausible* confusions within the same crop, the failure mode is rarely absurd, but is still actionable in the wrong direction. The system should be positioned as **decision support, not a replacement for expert judgement**, surface calibrated uncertainty, and make clear that liability for any intervention remains with the human user.

7. Conclusions

This work compared four progressively more sophisticated approaches to plant disease classification. Three findings are worth highlighting:

1. **Learned representations dominate handcrafted ones for fine-grained color-sensitive tasks** — the >24-point gap between V1 and the deep models is decisive.
2. **For saturated datasets, supervised fine-tuning of a pre-trained backbone is the highest-accuracy recipe** (V3, 99.87%), but the marginal gain over a well-designed custom CNN (V2, 99.66%) is small — and V3 reaches its peak in less than half the wall time.
3. **Self-supervised foundation models offer a strikingly favorable accuracy/cost trade-off** — DINOv3 frozen + linear probe reaches 98.35% with zero backbone training, making it the most attractive option for rapid prototyping, new-class onboarding, and edge deployment.

Future work should evaluate the same four pipelines on **in-field imagery** to quantify the domain-shift gap, and explore **partial fine-tuning of DINOv3's last transformer blocks** as a hybrid between the V3 and V4 recipes.